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EE3-02
EE9AC4

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2011

MSc and EEE PART III/IV: MEng, BEng.and ACGI

INSTRUMENTATION

Friday, 6 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : C. Papavassiliou
Second Marker(s) : T. Constandinou

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Special instructions for students

The following constants are given:

Electron charge: $e = 1.6 \times 10^{-19}\ \mathrm{C}$

Speed of light: $c = 3 \times 10^{8}\ \mathrm{m\,s^{-1}}$

Boltzmann constant: $k = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$

Standard temperature: $T = 290\ \mathrm{K}$

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The Questions

1. In this question we will investigate a directly sampled FM radio receiver.

An FM stereo radio channel has a bandwidth of 100 kHz centred on a carrier of frequency between 88 MHz and 108 MHz. Carriers are spaced 200 kHz apart.

You may assume that the entire FM band is populated at a power level of 100 fW per channel, and that there are no transmissions outside the FM band.

The receiver will use direct sampling, without preamplification. The noise bandwidth of the sampler is 216 MHz

a) Calculate the minimum bandpass sampling frequency which can be used to sample an FM channel without using an anti-aliasing filter.

[5]

b) Calculate the total noise power in one FM channel at room temperature if the minimum bandpass sampling rate you calculated in part (a) is used.

[5]

c) Calculate signal to noise ratio and, consequently, the number of bits of resolution possible if the signal sampled as described in part 1(a) is quantised by an ideal A/D converter.

[5]

e) Outline one technique which can be used to raise the resolution of the direct sampled converter to 12 bits. The sampling rate must not exceed 500MHz.

[5]

2.

a) Define the phase of a signal. What is the relation between phase difference and delay between two signals? Can the delay between two signals be uniquely determined if their phase difference is known?

[5]

b) Define phase noise in an oscillator, and discuss how it is related to jitter. Sketch the phase noise dependence on frequency offset from the nominal frequency of an oscillator. Discuss the effect that the phase noise of the local oscillator has on the performance of a heterodyne FM receiver.

[5]

c) With the aid of a diagram describe a counter-based frequency measurement instrument whose output is proportional to the unknown frequency of an input signal. Write an expression for the input frequency in terms of the counter value at the end of the measurement.

[5]

d) Discuss how clock jitter can enhance the performance of a counter-based frequency measurement instrument such as the one described in part 2(c) above.

[5]

3. The following devices are connected to form the instrument of figure 3:

A: Voltage multiplier.

B: Lead-lag low pass filter with a pole at τ_p and a zero at $\tau_z < \tau_p$

C: Monostable multivibrator which emits a pulse of a specified length τ every time its input crosses a threshold with a positive slope. The pulse duration is assumed to be very small.

D: Voltage controlled oscillator with gain K_0 = 1MHz/V

E: Second order band pass filter

Answer the following questions

a) Explain the function of each of the blocks A-E

[1 mark each]

b) Write an expression for the transfer function of each of the blocks A-E

[1 mark each]

c) Write an expression for the linearised closed loop gain of the circuit in figure 3 assuming that the input is a sinusoidal waveform and that the output is phase locked to the input. What is its steady state (s=0) response? What is the function of this instrument?

[5]

d) What is the name of this instrument? Comment on the stability of this instrument.

[5]

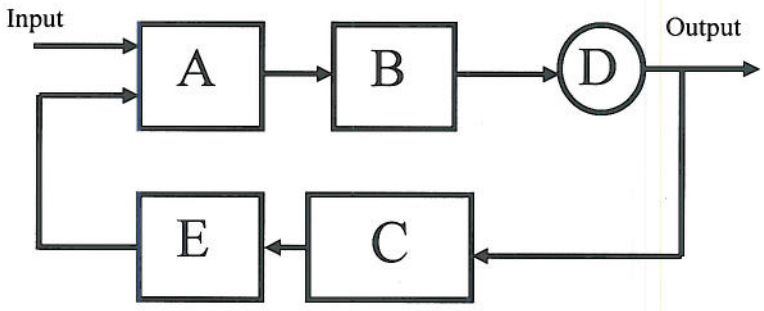


Figure 3: The instrument for question 3.

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4.

a) Draw a schematic diagram of a binary weighted digital-to-analogue converter (DAC). Discuss implementation and performance limitations of a binary weighted DAC.

[5]

b) Calculate the tolerance of the MSB output of a binary weighted DAC in terms of the circuit element values from the schematic you drew for part 4(a)

You may assume that the tolerance $T = \delta R / R$ of resistor values in an integrated circuit is a process dependent constant.

[5]

c) Calculate the ratio of the MSB (most significant bit) output tolerance to the LSB (least significant bit) output level. From this ratio derive an expression for the maximum number of bits feasible on a process which has resistor tolerance T. What is the converter number of bits for T=5% ?

[5]

d) Calculate the fractional tolerance in the value of a resistor composed of N equal resistors connected in parallel, if their individual tolerances are uncorrelated. Use this to show that thermometer coding can be used to construct a converter with twice the number of bits calculated in part 4(c) above.

You may assume that uncorrelated tolerances add like uncorrelated noise sources.

[5]

NOTE: You may ignore the internal resistance of any switches.

5.

a) Draw the block diagram of the successive approximation (SAR) analogue to digital converter. Describe the algorithm used by the SAR converter, and draw its state diagram for 3 bits.

[5]

b) Write an expression for the conversion time of a SAR A/D converter in terms of the clock frequency and the number of bits. Calculate the conversion time of a 24-bit SAR converter clocked at 10 MHz. What is the maximum signal bandwidth that can be applied to this converter?

[5]

c) Discuss one advantage and one disadvantage of a SAR converter when compared to a single slope converter. Why is the single slope converter becoming very popular in modern integrated circuit design?

[5]

d) Define "dither" and explain why dither is an essential ingredient in oversampled analogue to digital conversion.

[5]

6.

a) Define the Differential gain and the Common Mode Rejection Ratio (CMRR) of a difference amplifier.

[5]

b) Estimate the common mode gain of the amplifier in figure 6 if the op-amps are ideal and the resistor values are subject to a fractional tolerance $\delta = \delta R / R$.

Note: Tolerances add like uncorrelated noise sources.

[6]

c) Calculate the differential gain and the CMRR of the circuit in figure 6.

[4]

b) Modify the diagram of Figure 6 by adding three resistors, so that the common mode gain is increased by a factor of 10.

- What are the required values of the resistors?
- What is the value of the differential gain of the modified circuit?
- What is the value of the common mode gain of the modified circuit?

[5]

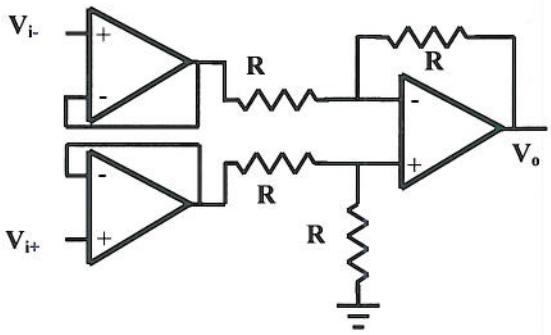


Figure 6: Circuit diagram for Question 6

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The Answers 2011

Question 1 [all computed new problem]

a) The sampling frequency must be at least $f_s > 2B = 40MHz$, and such that no harmonic mixing products overlap. The condition in the notes is that the upper end of the band is a harmonic of the sampling frequency. In this case the sampling frequency is twice the bandwidth. We can always pad the band with some extra bandwidth to an effective bandwidth B' in order to achieve this condition:

[3]

$$98MHz + \left(\frac{B'}{2}\right) = nB' \Rightarrow 98MHz = \left(n + \frac{1}{2}\right)B'$$

We investigate this for the largest n which results into a bandwidth $B' > 20MHz$
The largest n that satisfies the requirement is $n = 4$ and the minimum sampling rate is
$B' = 21.778MHz \Rightarrow f_s = 2B' = 43.556MHz$
A better answer observes that we can chose $f_s > 40MHz$, $f_s > 2f_u / n = 216/n \Rightarrow fs > 43.2MHz$

[2]

b) The noise bandwidth is 216Mz and is folded into a band equal to fs. So the noise folds in 5 times, and the total noise power in one channel is $P = 4kTB = 4kT \cdot 100kHz = 1.6fW$

[5]

c) The SNR is $SNR = 10\log(100) = 20dB$. Equate this with the quantisation noise of an n-bit converter to get: $20dB = 1.76 + 6.03n \Rightarrow n = 3$. This is a 3-bit radio!

[5]

d) Oversample with feedback. For instance, using a 2nd order $\Sigma\Delta$ at 216MHz we can get 3+4x2.5 =13 bits

[5]

Question 2:

a) [bookwork]

The **phase** of a signal is an index to the signal's value over one period, or for times between: $nT \le t \le (n+1)T \Rightarrow t = (n + \phi/2\pi)T$

Conventionally we measure phase in angle units (radians in 4.5). Delay is $\delta t = \frac{\varphi}{2\pi}T + nT$,

i.e. delay is not uniquely determinable from phase difference.

[5]

b) [bookwork+interpretation] Phase noise is random phase modulation of the oscillator output. The uncertainty in the timing of zero crossings is called jitter. Phase noise is equivalent to spurious FM modulation in the input signal, and therefore if the LO of a receiver has phase noise, the noise will appear at the output as input baseband noise.

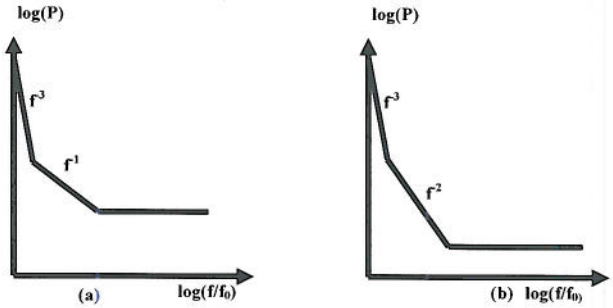
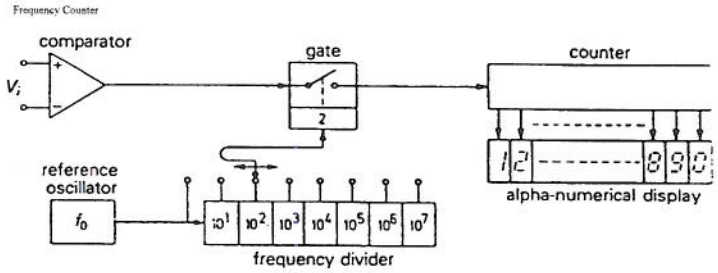


Figure Noise spectrum of oscillators around their output frequency. The oscillator is assumed to consist of an amplifier in a positive feedback loop. (a) High noise figure amplifier (b) Low noise figure amplifier.

[5]

c) [bookwork]



The input frequency is $N = \text{int}\left(\frac{\Delta T}{T}\right) = \text{int}\left(\frac{Df}{f_0}\right) \Rightarrow \frac{N}{D} f_0 < f < \frac{N+1}{D} f_0 \Rightarrow \lim_{D\to\infty} f = \frac{N+1/2}{D} f_0$ [5]

d) [interpretation]

This measurement is essentially a single slope A/D converter. Jitter will accelerate the convergence of the inequality, much like in an oversampled A/D converter.

[5]

Question 3.

a,b) [Extension of theory]

A: The multiplier acts as a Phase detector. Output voltage proportional to phase difference between inputs.

$V_o = K_d(\varphi_1 - \varphi_2)$.

[2]

B: The low pass filter filter: Smoothens the output of the phase detector.

$$H(s) = \frac{1+s\tau_z}{1+s\tau_p}$$

[2]

C: The monostable generates harmonics of the output frequency, i.e. acts as a frequency multiplier: $\varphi_{out} = N\varphi_{in}$

[2]

D: The Voltage controlled oscillator. Since the state variable is phase and not frequency, and phase is the integral of frequency, $\varphi_{out} = \frac{K_o}{s}V_{in}$

[2]

E: Second order band pass filter selects one of the harmonics:

$$H(s) = \frac{As/\omega_0}{(s/\omega_0)^2 + 2\zeta s/\omega_0 + 1}$$

[2]

c) [analysis, known problem]

$$\varphi_{out} = \frac{K_d H \frac{K_0}{s}}{1 + NK_d H \frac{K_0}{s}} = \frac{K_d H K_0}{s + NK_d H K_0}.$$

At s=0 this divides the phase (and the frequency) by N. If the steady state error is zero, then it maps the input phase to the same phase at a frequency equal to N times below the input frequency. The function of the instrument is to divide the frequency of the input waveform by a large integer number.

[5]

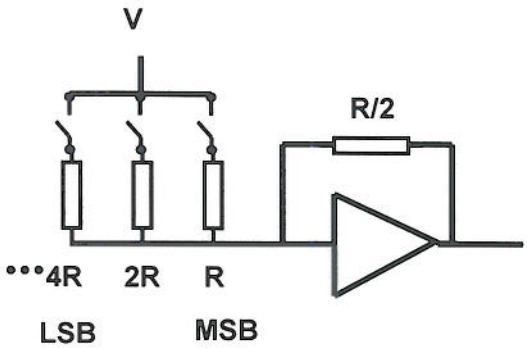
d) [bookwork]

Transfer Oscillator. It is difficult to stabilise due to the large loop gain, in part due to the phase multiplier in the loop.

[5]

Question 4.

a) [bookwork]



Binary weighted

Performance is limited by switches, and by op-amp noise, slew rate and bandwidth.

Implementation is limited by component tolerances. [5]

b) [new computed problem] $\frac{\delta R}{R} = T \Rightarrow |\delta I| = |V \delta G| = \left|\frac{V \delta R}{R^2}\right| = \frac{VT}{R_n} = \frac{VT}{2^{n-1}R}$

If the transimpedance is $R_T = R/2$, it follows that $|\delta V_n| = |\delta I_n| R_F = \frac{VT}{2^{n-1}R} R/2 = 2^{-n} VT$.

The variation in the MSB (n=0) is $\delta V_{MSB} = VT$ [5]

b) [new computed problem] The LSB output is the output for n=N $V_{LSB} = \frac{VR/2}{2^{N-1}R} = V/2^N$

The ratio of the uncertainty to half the LSB is $\frac{2\delta V_{MSB}}{V_{LSB}} = \frac{2VT}{V/2^N} = 2^{N+1} T \le 1 \Rightarrow N \le \log_2 \frac{1}{T} - 1$

T=0.05 N=3 [5]

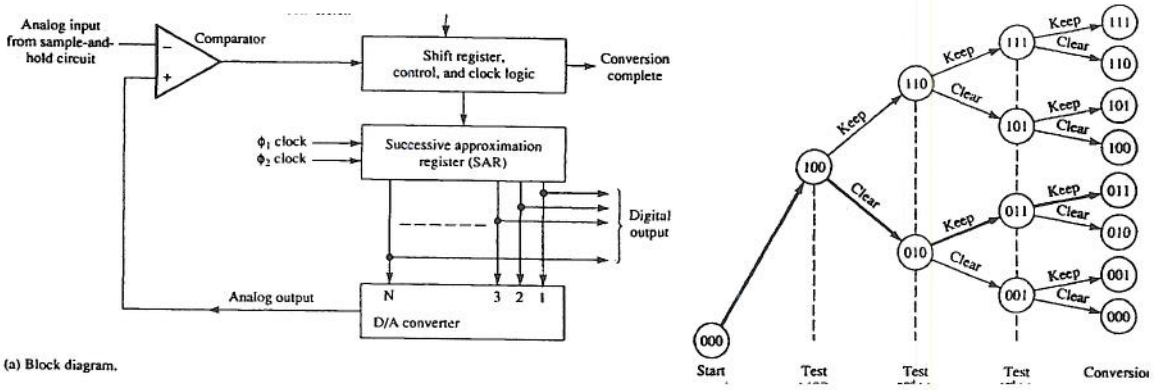
d) [new problem] A resistor of value R/N made of N resistors of R in parallel has a fractional tolerance:

$\frac{\delta(NR)}{NR} = \frac{\sqrt{N}\delta R}{NR} = \frac{1}{\sqrt{N}} \frac{R}{R} = \frac{T}{\sqrt{N}}$. If the resistors of value R are made out of N resistors of size NR in parallel, then the fractional tolerance of the MSB will be $\delta V_{MSB} = VT/\sqrt{2^{N-1}}$, while the value of the LSB is still the same. In a thermometer coded resistor, N resistors are connected to implement a resistor of value R/N. Therefore, the ratio of the uncertainty to half the LSB

$\frac{2\delta V_{MSB}}{V_{LSB}} = \frac{2VT/2^{\frac{N-1}{2}}}{V} = 2^{N/2+1/2} T \le 1 \Rightarrow N/2 \le \log_2 \frac{1}{T} - 1/2$

Which is over twice the width achievable by the binary weighted converter [5]

Question 5: [bookwork]



[5]

b) [interpretation]
This algorithm is logarithmic in time. The SAR of width N completes a conversion in N+1 cycles (seen also from the diagram above),
An 24 bit SAR at 10MHz needs 2.5usec for a conversion., As a result, this converter can make 400ks/sec and the maximum signal bandwidth is the Nyquist rate, 200kHz.

[5]

c) [interpretation]
The SAR has fast conversion and fxed conversion time as advantages relative to a single slope converter, which is slower and has a variable conversion time proportional to the input. However, it is limited by the comparator slew rate, and the accuracy of the D/A converter. The single slope converter times charging a capacitor to the analogue input. Extremely fast clocks are very easy to make in a digital technology.

[5]

d) [bookwork, interpretation]

To perform the averaging effectively in an oversampled converter we need to add some noise, to make sure that the signal and noise sum crosses frequently the converter's decision threshold. Such intentional noise is called **dither**. The required dither amplitude typically exceeds the converter's quantisation step. More precisely, we choose to represent the signal x by a random variable $y = x + e_x$ obtained by adding to the signal e_x, a random variable representing the dither noise. After N measurements, the ratio of the signal and dither noise averages is:

$$\left\langle \frac{\overline{x^2}}{\overline{e_x^2}} \right\rangle = \frac{N^2 x^2}{N\overline{e^2}} = N$$

[5]

Question 6:
a) [bookwork]

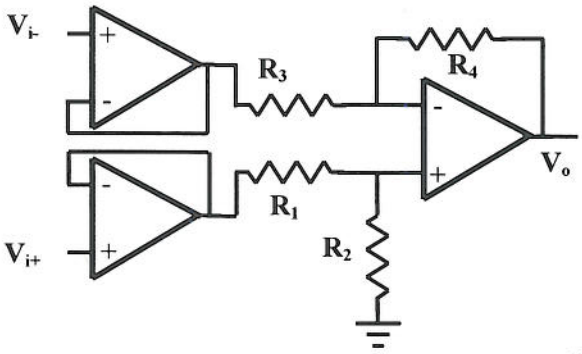
A difference amplifier has 2 inputs. The output is

$$V_o = G_{cm}\left(\frac{V_{i+}+V_{i-}}{2}\right)+G_{diff}\left(V_{i+}-V_{i-}\right)=V_{i+}\left(\frac{G_{CM}}{2}+G_{diff}\right)+V_{i-}\left(\frac{G_{CM}}{2}-G_{diff}\right)$$

This expression implicitly defines the common mode and differential gains.
The ratio of Common mode and differential gains is called the Common Mode Rejection Ratio (CMRR).

[5]

b) [computed example, partly known]



By straightforward circuit analysis, $V_o = V_{i+}\dfrac{R_2}{R_3}\dfrac{R_3+R_4}{R_1+R_2}-V_{i-}\dfrac{R_4}{R_3}=AV_{i+}+BV_{i-}$

Comparing to the definition of the gains, .

$A = G_{cm}/2+G_{diff}$, and $B = G_{cm}/2-G_{diff} \Rightarrow G_{cm}=A+B$, $G_{diff}=\dfrac{A-B}{2}$

By direct comparison with **Error! Reference source not found.** we conclude that the common mode gain is:

$$G_{cm}=\left(\frac{R_2}{R_3}\frac{R_3+R_4}{R_1+R_2}-\frac{R_4}{R_3}\right)=\frac{R_2R_3-R_4R_1}{R_3\left(R_1+R_2\right)}$$

If all the resistors are identical and subject to a fractional tolerance δ,

$$G_{cm}=\left(\frac{R_2}{R_3}\frac{R_3+R_4}{R_1+R_2}-\frac{R_4}{R_3}\right)=\left(1\pm\sqrt{2}\delta\right)\left(1\pm2\delta\right)-\left(1\pm\sqrt{2}\delta\right)\simeq\sqrt{8}\delta$$

[6]

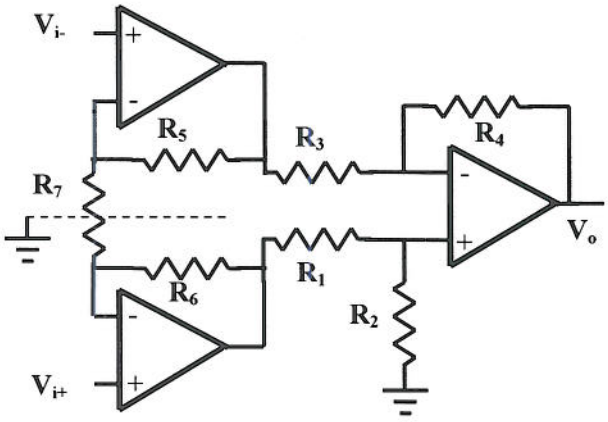
c) the differential gain is:

$$G_{diff}=\frac{1}{2}\left(\frac{R_2}{R_3}\frac{R_3+R_4}{R_1+R_2}+\frac{R_4}{R_3}\right)=\frac{1}{2R_3}\frac{R_2\left(R_3+R_4\right)+R_4\left(R_1+R_2\right)}{\left(R_1+R_2\right)}=1$$

The CMRR is $CMRR = \frac{G_{diff}}{G_{cm}} = \frac{1}{\sqrt{8\delta}}$ [4]

d) [bookwork-interpretation]

The CMRR can be enhanced by using the 3 op-amp standard difference amplifier:



Both the differential gain and the CMRR will be increased by a factor of

$\delta G = 1 + \frac{2R_5}{R_7} = 10 = G_{diff}$. The common mode gain will remain constant.

[5]

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